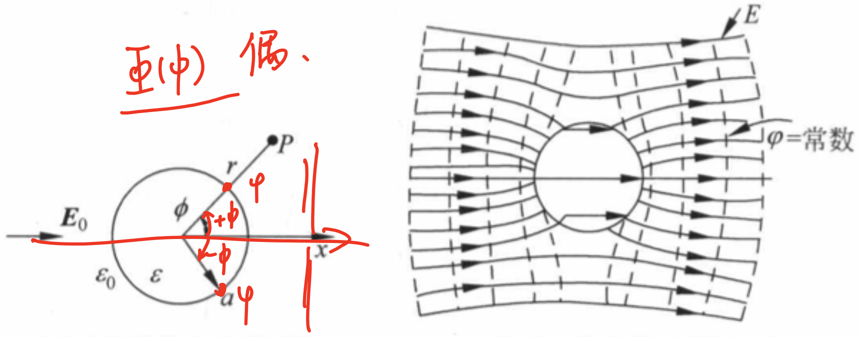
例题 4-9 若在电场强度为 $\boldsymbol{E}_0$ 的均匀静电场中放入一个半径为 a 的电介质圆柱，柱的轴线与电场互相垂直，介质柱的介电常数为 ε，柱外为真空，如图 4-9(a)所示，求柱内、柱外的电场。

$\Phi(\phi)$ 偶.



(a) 电场中的电介质圆柱　　(b) 柱内、外电位及电场分布

图 4-9　电介质圆柱的电场分布

$\nabla^2\varphi = 0$

$$\nabla^2\varphi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\varphi}{\partial\phi^2} + \cancel{\frac{\partial^2\varphi}{\partial z^2}}$$

$$\begin{cases} \varphi_1\big|_{r=0} = 0 \\ \varphi_2\big|_{r=\infty} = -E_0 r\cos\phi \\ \varphi_1 = \varphi_2\big|_{r=a} \ \checkmark \\ \varepsilon\dfrac{\partial\varphi_1}{\partial r} = \varepsilon_0\dfrac{\partial\varphi_2}{\partial r}\Big|_{r=a} \end{cases}$$

$$\nabla^2\varphi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\varphi}{\partial\phi^2}$$

令 $\varphi(r,\phi) = R(r)\cdot\Phi(\phi)$

$$\frac{1}{r}\frac{\partial}{\partial r}(rR'\Phi) + \frac{1}{r^2}R\Phi'' = 0$$

$$r\frac{\partial}{\partial r}(rR'\Phi) + R\Phi'' = 0$$

$$r[R'\Phi + rR''\Phi] + R\Phi'' = 0$$

$R\Phi$

$$\therefore r^2R''\Phi + rR'\Phi + R\Phi'' = 0$$

$$\frac{r^2R''+rR'}{R} + \frac{\Phi''}{\Phi} = 0$$

$$\frac{r^2R''+rR'}{R} = -\frac{\Phi''}{\Phi} = n^2$$

$\{\ r^2R'' + rR' - n^2R = 0$

$$\Phi'' + n^2\Phi = 0 \quad \checkmark$$

先解 $r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} - n^2R = 0$

令 $r = e^t$ $\quad t = \ln r$

$$\frac{dR}{dr} = \frac{dR}{dt}\cdot\frac{dt}{dr} = \frac{1}{r}\frac{dR}{dt}$$

$$\frac{d^2R}{dr^2} = \frac{d}{dr}\left(\frac{1}{r}\frac{dR}{dt}\right)$$

$$= -\frac{1}{r^2}\frac{dR}{dt} + \frac{1}{r}\frac{d^2R}{dt^2}\cdot\frac{dt}{dr}$$

$$= -\frac{1}{r^2}\frac{dR}{dt} + \frac{1}{r^2}\frac{d^2R}{dt^2}$$

$$r^2\left(-\frac{1}{r^2}\frac{dR}{dt} + \frac{1}{r^2}\frac{d^2R}{dt^2}\right) + r\left(\frac{1}{r}\frac{dR}{dt}\right) - n^2R = 0$$

$$-\frac{dR}{dt} + \frac{d^2R}{dt^2} + \frac{dR}{dt} - n^2R = 0$$

$$\frac{d^2R}{dt^2} - n^2R = 0$$

$$R(t) = Ae^{nt} + Be^{-nt}$$

$$= Ar^n + Br^{-n}$$

再解 $\Phi'' + n^2\Phi = 0$

$$\Phi(\phi) = C\cos(n\phi) + D\sin(n\phi)$$

$$\varphi(x,y) = \sum_{n=1}^{\infty}\left(A_n r^n + B_n r^{-n}\right)\left(C_n\cos(n\phi) + D_n\sin(n\phi)\right)$$

$$= \sum_{n=1}^{\infty} r^n\left(A_n\cos(n\phi)\right) + r^{-n}\left(C_n\cos(n\phi)\right) \quad \times$$

① 球内

$$\boxed{\varphi_1 = \sum_{n=1}^{\infty} A_n r^n \cos(n\phi)}$$

$r\to\infty$ $\to 0$

② 球外

$$\varphi_2 = \sum_{n=1}^{\infty} r^n\left(A_n\cos(n\phi)\right) + r^{-n}\left(C_n\cos(n\phi)\right) = -E_0 r\cos\phi$$

$r A_1 \cos\phi + \cancel{r^2 A_2 \cos(2\phi) + \ldots}$

$$\varphi_2 = -E_0 r\cos\phi + \sum_{n=1}^{\infty} C_n r^{-n} \cos(n\phi)$$

$$\sum_{n=1}^{\infty} A_n a^n \cos(n\phi) = -E_0 a\cos\phi + \sum_{n=1}^{\infty} C_n a^{-n} \cos(n\phi)$$

$$A_1 a \cos\phi + \cancel{A_2 a^2 \cos(2\phi)} + \cancel{A_3 a^3 \cos 3\phi} + \ldots = -E_0 a\cos\phi + C_1 a^{-1}\cos\phi + \cancel{C_2 a^{-2}\cos 2\phi} + \cancel{C_3 a^{-3}\cos 3\phi}$$

$$A_1 a = -E_0 a + C_1 a^{-1} \rightarrow \boxed{A_1 = -E_0 + C_1 a^{-2}}$$

$$A_2 \boxed{a^2} = C_2 \boxed{a^{-2}} \rightarrow \cancel{A_2 = C_2 = 0}$$

$$A_3 a^3 = C_3 a^{-3} \rightarrow \cancel{A_3 = C_3 = 0}$$

$n=1$

$$\varepsilon \frac{\partial \varphi_1}{\partial r} = \varepsilon_0 \frac{\partial \varphi_2}{\partial r} \Big|_{r=a}$$

$$\varphi_1 = \sum_{n=1}^{\infty} A_n r^n \cos(n\phi) \qquad \frac{\partial \varphi_1}{\partial r} = \sum_{n=1}^{\infty} A_n n a^{n-1} \cos(n\phi)$$

$$\varphi_2 = -E_0 r\cos\phi + \sum_{n=1}^{\infty} C_n r^{-n} \cos(n\phi) \qquad \frac{\partial \varphi_2}{\partial r} = -E_0\cos\phi + \sum_{n=1}^{\infty} (-n) C_n a^{-n-1} \cos(n\phi)$$

$$\varepsilon\left(A_1 \cdot 1 \cdot a^{1-1} \cancel{\cos\phi}\right) = \left(-E_0 \cancel{\cos\phi} + (-1) C_1 a^{-2} \cancel{\cos(\phi)}\right)\varepsilon_0$$

$$\boxed{\varepsilon A_1 = (-E_0 - C_1 a^{-2})\varepsilon_0}$$

$$\varepsilon(-E_0 + C_1 a^{-2}) = \varepsilon_0(-E_0 - C_1 a^{-2})$$

$$\therefore -\varepsilon E_0 + \varepsilon C_1 a^{-2} = -\varepsilon_0 E_0 - \varepsilon_0 C_1 a^{-2}$$

$$\therefore (\varepsilon a^{-2} + \varepsilon_0 a^{-2}) C_1 = (\varepsilon - \varepsilon_0) E_0$$

$$C_1 = \frac{\varepsilon - \varepsilon_0}{\varepsilon a^{-2} + \varepsilon_0 a^{-2}} E_0$$

$$C_1 = \frac{\varepsilon - \varepsilon_0}{\varepsilon + \varepsilon_0} a^2 E_0 .$$

$$A_1 = -E_0 + C_1 a^{-2}$$

$$= -E_0 + \frac{\varepsilon - \varepsilon_0}{\varepsilon + \varepsilon_0} \cancel{a^2} E_0 \cdot \cancel{a^{-2}}$$

$$= \frac{\varepsilon - \varepsilon_0 - (\varepsilon + \varepsilon_0)}{\varepsilon + \varepsilon_0} E_0$$

$$A_1 = \frac{-2\varepsilon_0}{\varepsilon + \varepsilon_0} E_0$$

$$\begin{cases} \varphi_1 = A_1 r \cdot \cos\phi = \dfrac{-2\varepsilon_0 E_0}{\varepsilon + \varepsilon_0} r\cos\phi & r\text{内} \\ \varphi_2 = -E_0 r\cos\phi + \dfrac{\varepsilon - \varepsilon_0}{\varepsilon + \varepsilon_0} a^2 E_0 \dfrac{1}{r} \cos\phi & r\text{外} \end{cases}$$